

IFFCO Paradeep Phosphogypsum – Production and Context

IFFCO's Paradeep (Odisha) complex produces ~875,000 MTPA of phosphoric acid [4†L329-L338] . (See image below.) This wet-acid process yields **phosphogypsum** (PG) as a byproduct at roughly a 4–5:1 mass ratio to acid [35†L149-L158] . In practice Paradeep generates on the order of **40 lakh tonnes per year** (≈ 4 million t/yr) of PG [30†L142-L150] . The IFFCO site uses lined “stacks” and ponds to handle the slurry – PG is pumped as a 10–15% solids slurry to large lined storage areas [30†L142-L150] . Daily output is on the order of ~11,000 t ($\approx 333,000$ t/month) of wet gypsum.

[58†embed_image] Figure: IFFCO Paradeep fertilizer complex (Odisha). This unit makes ~8.75 lakh t/yr phosphoric acid [4†L329-L338] , generating roughly 40 lakh t/yr of phosphogypsum [30†L142-L150] .

Composition/Properties: Phosphogypsum is primarily **calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)** [40†L130-L139] . Unlike pure gypsum, it contains impurities: residual phosphate and sulfuric acid, fluorides, organic matter, heavy metals, and trace radionuclides (U, Th, Ra from the phosphate rock) [40†L130-L139] [35†L169-L178] . It is typically a fine, wet powder (often 6–20% moisture [10†L117-L120]) with mildly acidic filtrate (pH ~2.5–3). Moisture is generally <15% for cement use [10†L117-L120] . Because of the natural radioactivity, PG is classified as “TENORM” (Technologically Enhanced Naturally Occurring Radioactive Material) in some countries [35†L169-L178] [40†L130-L139] . These impurities mean PG must often be treated or blended before use.

Uses of PG (Applications): Despite being a waste, PG has multiple potential uses **if processed properly**. Key fields include:

- **Cement and construction materials:** Gypsum is added (4–5%) to Portland cement as a set regulator. Fertilizer-industry PG can substitute mined gypsum in cement, concrete, bricks, plasterboard, etc [38†L281-L289] [10†L46-L53] . For example, UltraTech Cement is **sourcing IFFCO PG for cement** manufacture [8†L178-L185] [10†L46-L53] . Studies show PG can improve road base and brick properties when properly blended [38†L281-L289] . Cement plants in India needed ~13–17 Mt of gypsum in 2020-21 (for ~334 Mt cement) [10†L46-L53] , creating strong demand for PG alternatives.
- **Soil amendment (Agriculture):** Gypsum is a well-known soil conditioner and source of Ca and S [38†L258-L265] . PG has been trialed as a fertilizer amendment (especially on acidic or sodic soils) because it can improve soil structure and supply nutrients [38†L258-L265] . (E.g. China studies report increased yields when used correctly [33†L131-L139] .) In India, gypsum is a plant nutrient (calcium/sulfur source) and PG could serve similarly if contaminants are managed.
- **Other industrial uses:** Proposed applications include mine/land reclamation fill, road sub-base and aggregate, stucco/plaster, and conversion into products like ammonium sulfate or gypsum boards

【38†L281-L289】 【38†L291-L293】 . For example, fertilizer.org notes PG can be used for “gypsum boards, road base materials, and agricultural soil amendments” 【33†L134-L142】 【38†L281-L289】 . In practice these uses are limited by impurity/radioactivity concerns, but remain research and pilot areas.

- **Cement industry byproducts:** The cement industry is explicitly pursuing “*chemical gypsum*” alternatives. The Cement Manufacturers’ Assoc. (CMA) notes India’s PG output (~8.46 Mt in 2020-21 【10†L95-L103】) and suggests cement plants can consume many million tonnes of PG per year. They recommend preprocessing PG to meet quality standards (e.g. <0.25% soluble P_2O_5 , <15% moisture) 【10†L113-L121】 . CMA also notes that most bulk gypsum transport is by rail, but current freight classification (gypsum Class 150) makes it costly 【10†L125-L134】 .

IFFCO PG Stockpile & Management: Because PG is legally a “waste” (requiring secure disposal), IFFCO has built massive storage stacks. Engineers report that Paradeep needed ~250 acres of lined stack for 15 years of PG (~60 million t) 【30†L142-L150】 . In fact, IFFCO contracted new stack phases (e.g. a ₹190 crore project in 2016 for 60% of a new stack 【51†L109-L118】). PG slurry settles in these ponds; clear acidic decant is recycled back to the plant 【30†L142-L150】 . The stockpiles are tens of meters high. Disposal is **environmentally sensitive**: leachate is acidic and PG can release radon and heavy metals into groundwater/soil 【33†L123-L130】 【40†L130-L139】 . This “dump” adjacent to the plant is therefore a major liability and driver of reuse efforts.

Current Reuse Initiatives: To reduce the stockpile, IFFCO is **selling and giving away PG** to end-users and collaborating on reuse:

- **Cement sales:** IFFCO has supplied PG to Odisha and Gujarat cement plants. Notably, UltraTech Cement arranged to **transport 57,000 t of PG** from Paradeep to its Gujarat works in 2023 【8†L161-L169】 【45†L149-L158】 . UltraTech sourced this PG from IFFCO (and Paradeep Phosphates Ltd.) 【8†L178-L185】 【45†L149-L158】 as a partial gypsum substitute in cement. Other local cement firms (e.g. OCL India’s Odisha plant) and Odisha Agro are known PG buyers 【41†L1725-L1733】 . Reports say IFFCO sold ~5.09 lakh t in 2011-12 to cement and other users 【41†L1725-L1733】 .
- **Stack expansion:** Recognizing limits to recycling, IFFCO is expanding its own ponds (see above). The 2016 Essar contract shows continued investment in **new PG stacking capacity** 【51†L109-L118】 .
- **Policy push:** The Indian government and industry view PG reuse as circular economy. The DPIIT and CMA emphasize diverting PG from landfill 【8†L190-L199】 【45†L187-L191】 . (For example, using PG instead of imported gypsum “saves precious forex” 【45†L179-L185】 .) There are proposals for road/landfill projects using PG, though these are in early stages.

Potential Buyers and Fields: The **cement industry** is by far the largest potential consumer. UltraTech (Aditya Birla Group) has led by example 【8†L178-L185】 . Other major cement groups (ACC/Ambuja, Shree Cements, Dalmia, etc.) could similarly use millions of tonnes. CMA data imply ~13–17 Mt gypsum needed for cement now 【10†L46-L53】 , vs ~8–9 Mt PG available 【10†L95-L103】 , leaving ample scope. Beyond cement, large building-materials firms (gypsum board/plaster producers) could use PG if allowed. On the agricultural side, state agriculture departments or large farm co-ops might use PG as a soil amendment (especially in alkaline or sulfur-poor regions) 【38†L258-L265】 . Government road projects might also be candidates (some foreign regulators have approved PG for road base).

Transportation & Costs: Moving PG in bulk is expensive. CMA notes rail is preferred, but gypsum's freight class (150) makes it pricey 【10†L125-L134】. By road, truck rates average only ~₹3.9–4.0 per ton-km in India 【18†L446-L454】. Thus transporting 1 t over 500 km costs ~₹2,000; 1000 km ~₹4,000 (about \$45). By contrast, rail can be cheaper per km (though exact rates depend on class). For example, UltraTech's coastal shipment leveraged waterways (cheaper per tonne) 【8†L161-L169】.

Because *gypsum is bulky and low-value*, the cost of transport often dominates its delivered price. (CMA suggests lowering gypsum's freight class to cut landed cost 【10†L125-L134】.) In India, imported natural gypsum often costs on the order of US\$17–33/ton (~₹1,400–2,500/t) 【66†L1-L4】. IFFCO's PG, being a waste byproduct, is typically **much cheaper**: it is often provided at only a token charge or even free (users usually pay shipping only). UltraTech notes replacing imported gypsum with PG "saves precious forex" 【45†L179-L185】. In practice, cement makers effectively get PG at well below the market price of mined gypsum, making it a very low-cost gypsum source once transportation is arranged.

Production & Pricing Summary: In sum, IFFCO-Paradeep's PG output is huge (~40 lakh t/yr 【30†L142-L150】). This overhang has motivated storage expansion and budding reuse deals. IFFCO has facilitated large sales (e.g. to cement plants) and logistical innovations (waterway shipping 【8†L161-L169】). The industry outlook is that virtually all PG will need to find productive use: cement demand is vast (cement output ~400 Mt in FY23 【45†L171-L174】) and even FGD gypsum (power plant) is being tapped. If used in cement/clinker, PG can displace imported gypsum, cutting costs. IFFCO likely "sells" PG at only a handling fee, so its effective price to buyers is far below that of mined gypsum (which is typically ₹2,000–3,000+/t). Thus PG can be 50–100% cheaper on a per-tonne basis, though the true cost depends heavily on transport distance.

References: Authoritative reports and industry sources (IFFCO documents, Fertiliser Assoc. data, CMA, UltraTech and news releases, etc.) were used. Specific data and statements above are cited accordingly.
